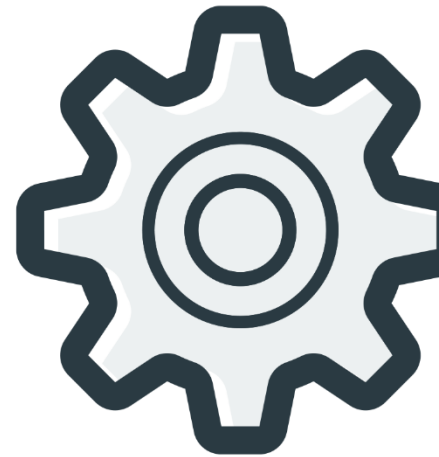
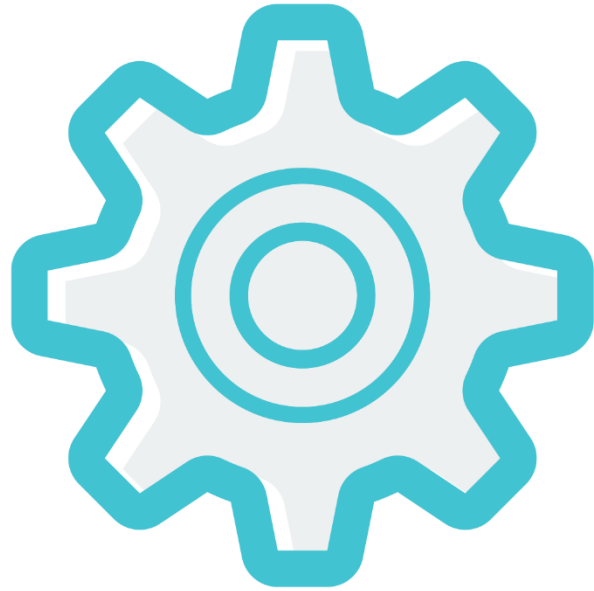


JUNIOR ROBOTICS

MANUAL

PATH FINDING ROBOT



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WARNING: The kit contents are not suitable for children under 3 years. Choking hazard – small parts may be swallowed or inhaled.

Store the kit materials and assembled models out of reach of small children.



HANDLING AND PRECAUTIONS

General:

- Assembly and operation of this kit should be done only under the supervision of an adult
- All parts of this kit should be stored and operated away from water and fire sources
- All parts of this kit should never be put into the mouth
- Throwing or dropping the parts may cause damage to the parts

Building/Operating:

- All the tools for assembling must be used with extreme precaution
- The parts should not be forcibly bent, attached or removed
- The parts should not be touched when in motion
- The power supply, motors, sensors and other components should be checked before operating
- The power supply should be turned off when connecting the Arduino to the computer for programming

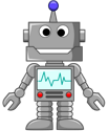


Batteries:

- The batteries should not be inserted in the battery holder before mounting and connecting the battery holder onto the model
- The batteries should be inserted correctly with the (+) and (-) in the correct direction
- Always use the provided batteries or batteries of correct size and rating (3.7V)
- New and used batteries should not be mixed for operation
- Always close the battery holder with the lid after removal or insertion of batteries

Workshop 1

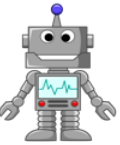
- Adding Liquid Crystal Display



Hi Andy, welcome to next part of the course which is to make a path finding robot. First, we start with adding an LCD display



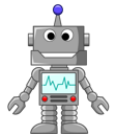
That is exciting, what is LCD display?



Liquid Crystal Display also known as LCD is where you can display letters and numbers on the screen. LCD is extremely useful when you need to troubleshoot your robot. We are using a 16 x 2 Display, which means it has 16 columns and 2 rows. we can set the cursor where we want to



Okay, what is the first step?



First step is to assemble the display!

Step 1

Place two golden standoffs in the holes near the battery holder using the screws from the bottom of the robot frame

Refer to figures 1.02 and 1.03



Figure 1.01



Figure 1.02



Figure 1.03

Refer to the images 1.04 and 1.05 to assemble the LCD display

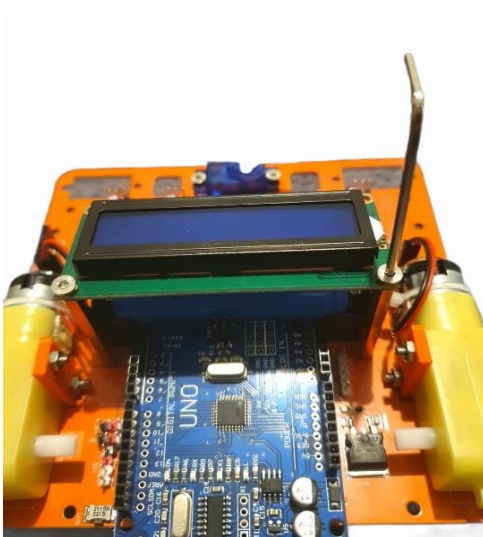


Figure 1.04



Figure 1.05

Step 2

Connections:

Connect the LCD to the LCD pins highlighted in the board, make sure pins are connected correctly

- i) GND: ground or --
- ii) VCC: +
- iii) SDA - SDA
- iv) SCL - SCL

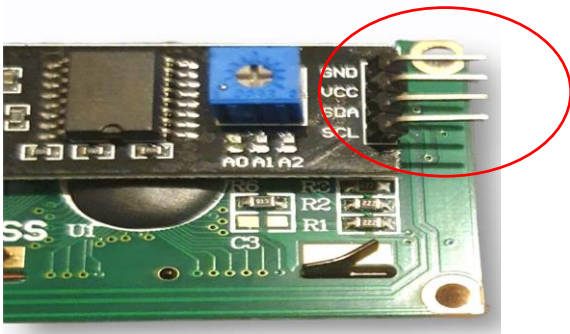


Figure 1.06

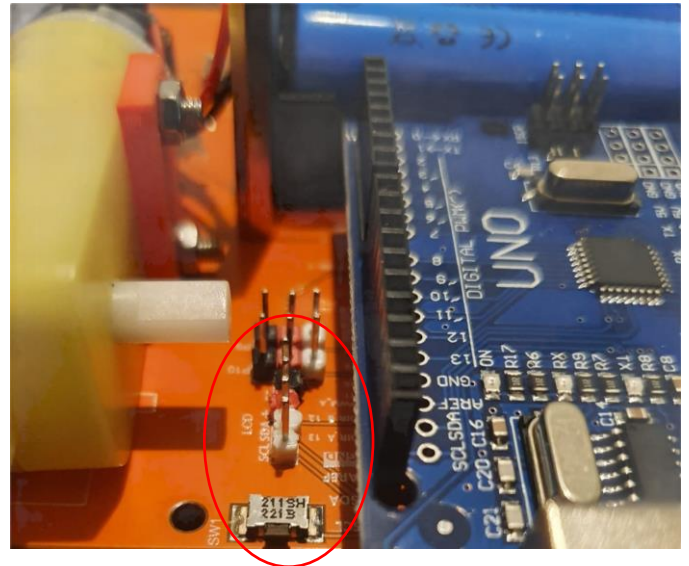


Figure 1.07

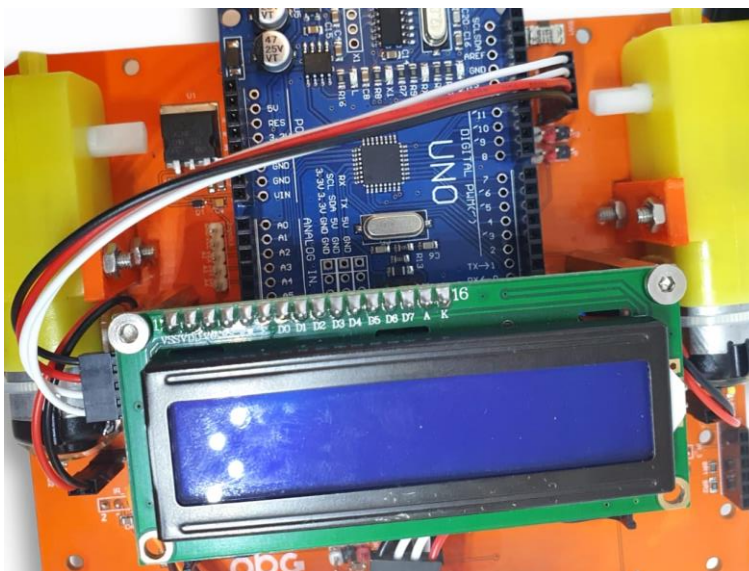


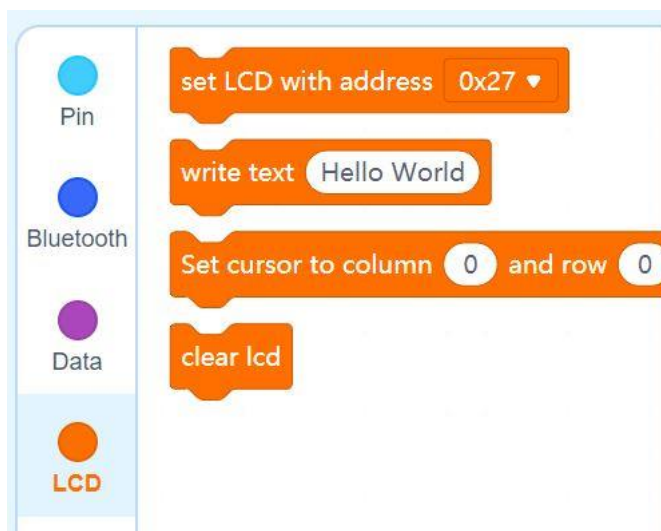
Figure 1.08

Step 3

After installation, we are ready to program. The first and important parameter is the address of the LCD

Address – The address represents the address of the LCD, we need this as we are using I2C protocol. There are two options available, 0x27 and 0x3F, try both the addresses and one of them will work.

We can use the LCD module as shown in the below figure to set up the address, column and writing text or displaying numbers



Let’s make a program for our LCD to print “Hello World”



Figure 1.10

Challenge 1

Display your first name in the first row and last name in the second row

Workshop 2

Let us make a counter to count from 1 to 100 on LCD and display the same. We need to create a variable for this. Variable is like an empty container in which we can store different letters or numbers

Refer to figures 1.11 and 1.12 on selecting the variables module in the mblock and declaring a variable to use in our counter program

New Variable



New variable name:

Variable_count

☒ For all sprites

☐ For this sprite only

Cancel

OK

Figure 1.12

when Arduino Uno starts up

set LCD with address 0x27

set Variable_Count to 0

repeat 100

Set cursor to column 0 and row 0

write text Variable_Count

wait 1 seconds

clear lcd

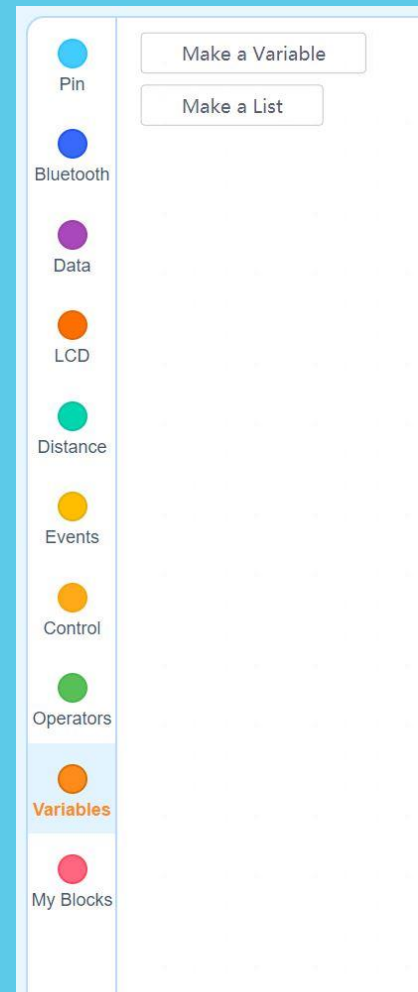


Figure 1.11

Figure 1.13

Workshop 3

- Adding Servo Motor



Hi Aebo, what are we doing today?



Andy, we are learning about how to use our servo motor



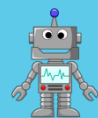
What is a servo motor?



A servomotor is a linear actuator that allows for precise control of angular or linear position which uses position feedback to control our robot's motion and final position



Are we adding the servo motor to our robot today?



Yes, let's start today's workshop



Step 1

First, take a servo motor and a servo horn. The servo has three cables:

- i) Brown → GND
- ii) Red → Vcc
- iii) Orange → Out

Step 2

Insert the servo horn with four sides on top of the motor and insert the motor into the board with the cable going in first and connect it to the Servo(pin8) port as shown in the figure. The side with the brown cable goes into the negative side of the servo motor pin located in the board

Refer to the following images for assembling the servo motor



Figure 1.15

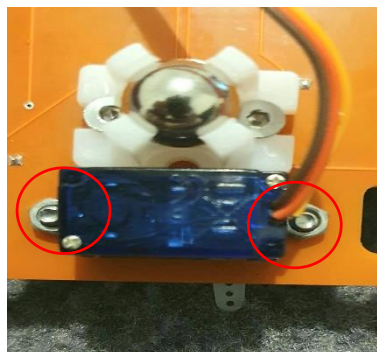


Figure 1.16

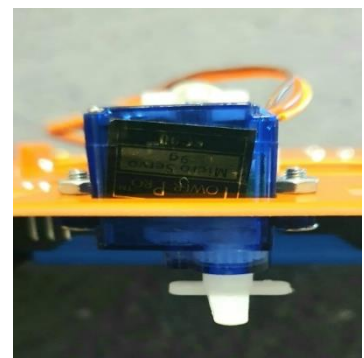


Figure 1.17

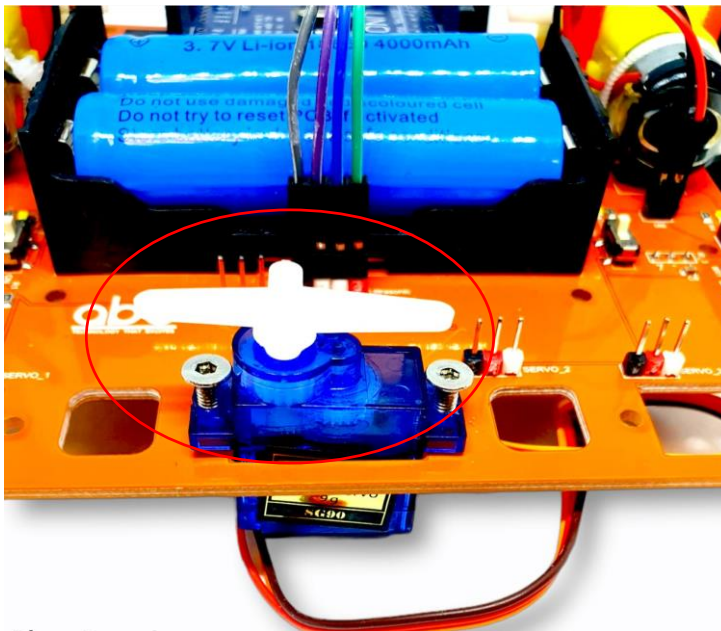


Figure 1.18

Step 3

To move the servo, first, upload the code in Figure 1.19 into the robot. This code makes the servo move between angle 0 degrees and 180 degrees with the delay of 1 second

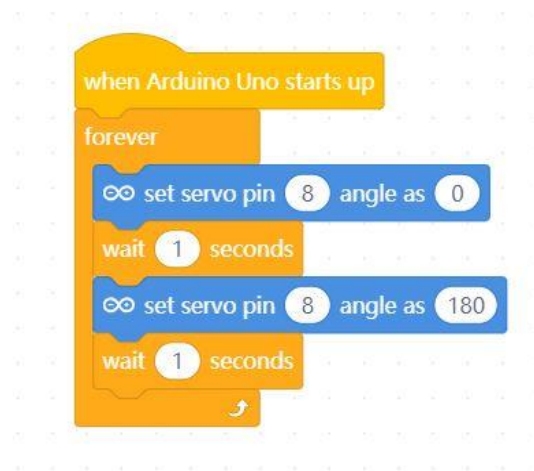


Figure 1.19

Challenge 1

To move servo horn from 0 to 180 degrees in interval of one degree

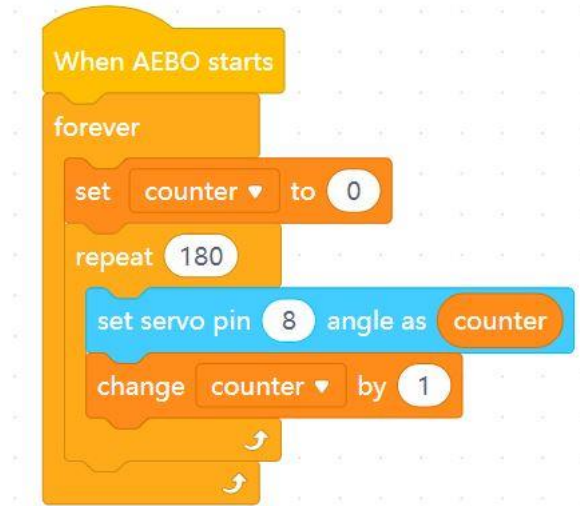


Figure 1.20

Challenge 2

To move servo horn from 0 to 180 and 180 to 0 degrees in interval of one degree

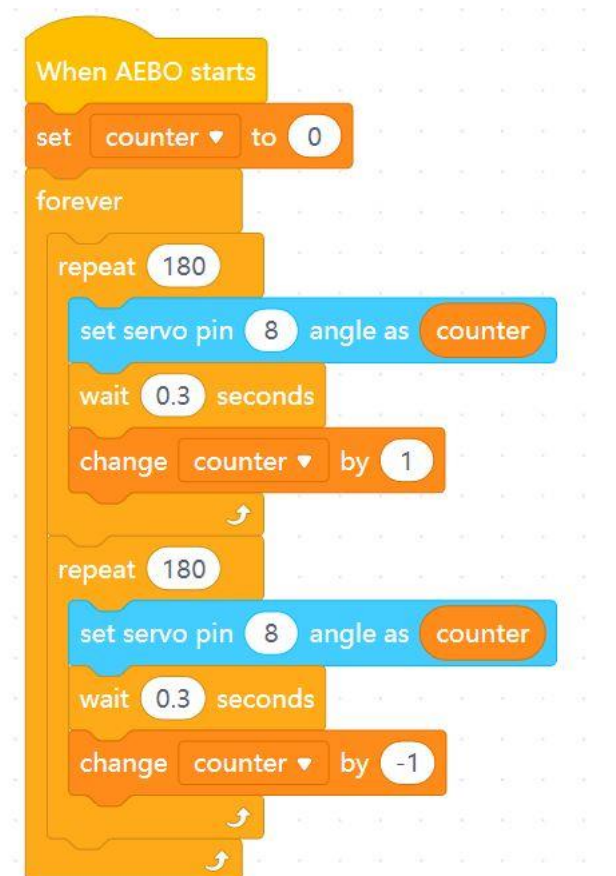


Figure 1.21

Workshop 4

- Adding Ultrasonic sensor



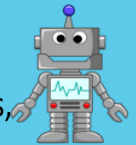
Hi Aebo, what are we doing today?

Andy, we are learning how to use our ultrasonic sensor



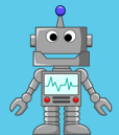
What is an ultrasonic sensor?

Ultrasonic sensors are devices that generate or sense ultrasound energy, it measures the distance of a target object by emitting ultrasonic sound waves, and converts the reflected sound into an electrical signal



Are we connecting the sensor to our Aebo?

Yes, let's start



Step 1

Insert the ultrasonic sensor into the casing

The next step is to connect the ultrasonic sensor with the mounting. Before doing this, upload the servo calibration program into your Arduino. This to make sure the servo motor is calibrated to face forward when the servo angle is 90



Figure 1.22



Figure 1.23



Figure 1.24

As shown in the below figures, connect the servo horn to the ultrasonic sensor mount and secure it using the screws inside the servo motor packet and connect the pins of ultrasonic sensor to the pins on the board using jumper cables

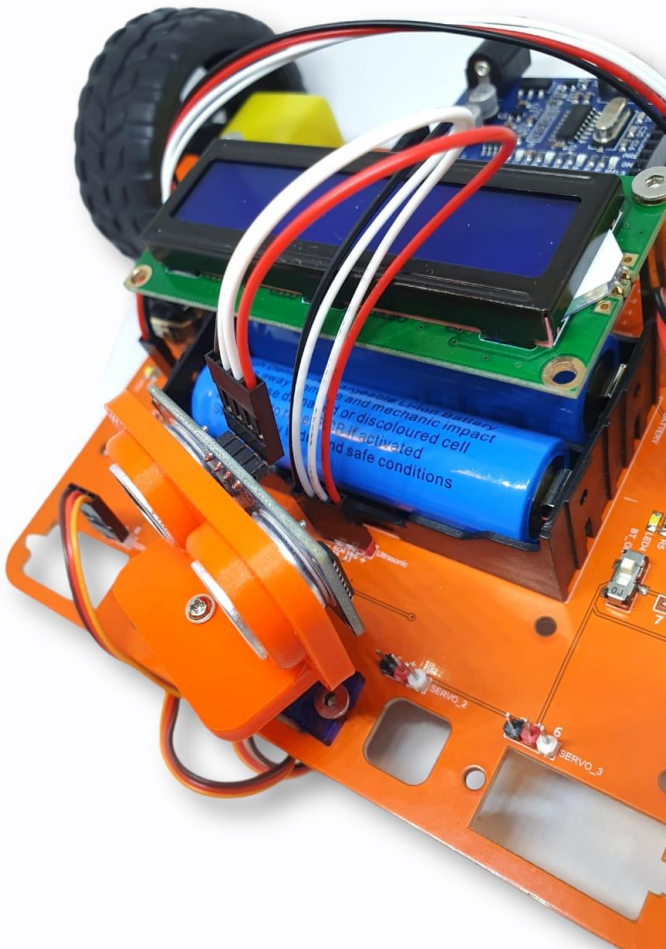


Figure 1.25



Figure 1.26



Figure 1.27

To get the value of the distance of the obstacle in front of the robot, refer to the code in figure 1.29, the distance is displayed in centimetres

Go to the distance module and click on the read ultra-sonic sensor to get the distance readings for our program

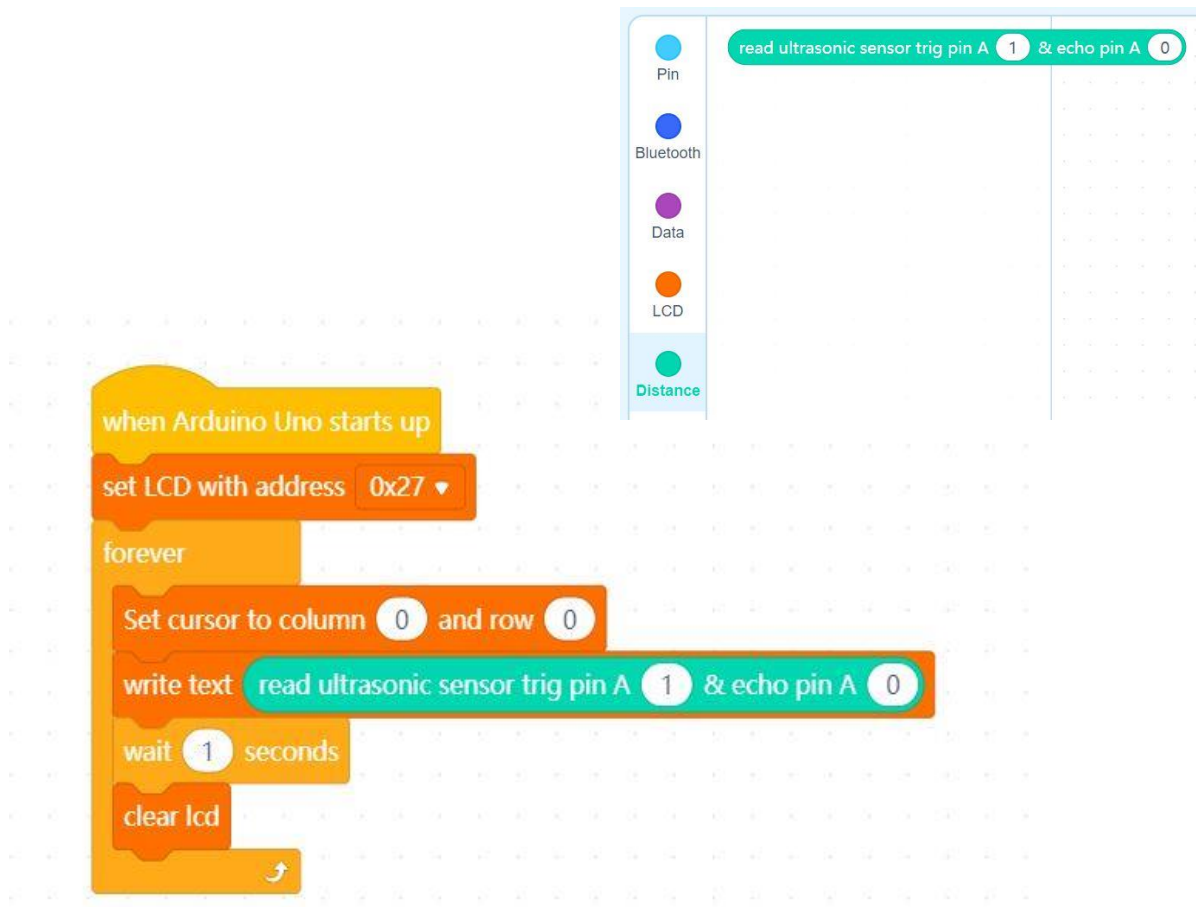


Figure 1.29

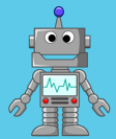
Workshop 5 & 6

- Social Distancing Robot



Hi Aebo, what are we doing today?

Andy, we are making our Aebo a social distancing Robot



What is a social distancing robot?

A social distancing robot will maintain safe distance from any obstacle, if an object comes near the robot, the robot moves backward, if the object moves away from the robot, then robot moves forward



Sounds good, let's start coding then

We can use earlier defined blocks of forward, backward and stop_robot in our program

Final code for social distancing robot

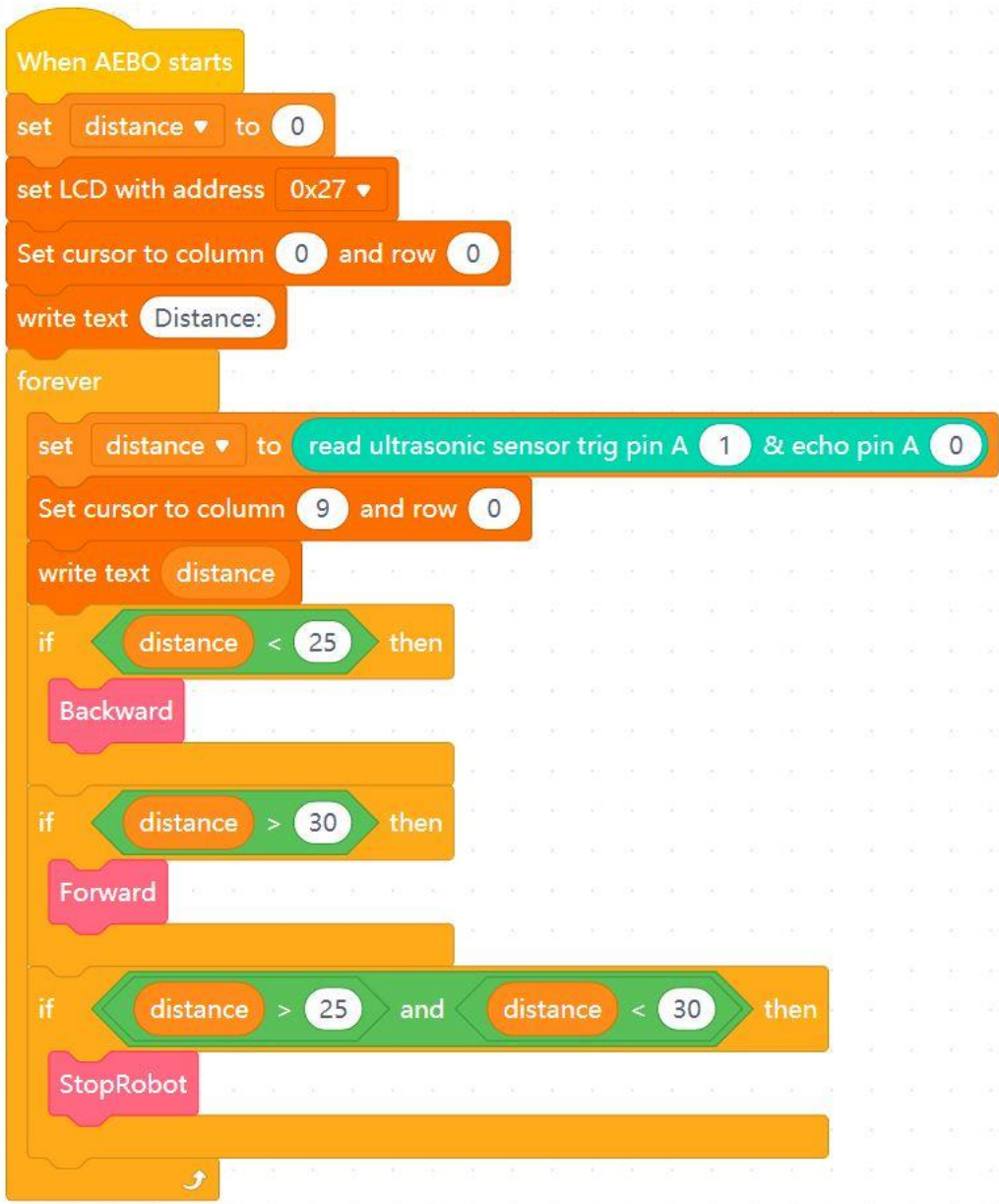


Figure 1.30

define Forward

```

set digital pin 12 output as High
set digital pin 13 output as High
set speed pin 3 output as 100
set speed pin 11 output as 100
  
```

define Backward

```

set digital pin 12 output as Low
set digital pin 13 output as Low
set speed pin 3 output as 100
set speed pin 11 output as 100
  
```

define StopRobot

```

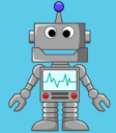
set digital pin 12 output as High
set digital pin 13 output as High
set speed pin 3 output as 0
set speed pin 11 output as 0
  
```

Workshop 7 & 8

- Path finding Robot



Hi Aebo, what are we doing today?

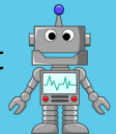


Andy, we are making a path finding robot



What is a path finding robot?

A path finding robot travels through an obstacle course where humans cannot enter, for example robots used in mining, rescue robots used in underground caves



What are the steps to be followed?

We will learn the steps in detail along the way let's start



Follow the below steps to complete the path finding robot

- We need to program our robot so it moves forward when the distance in front of the robot is more than 20cms.
- If distance is less than 20cms, robot looks left and right.
- Gets the distance of each side.

Compares the distance and travel towards the direction which has greater distance

Refer to figures 1.31 and 1.32 for the final code for path finding robot

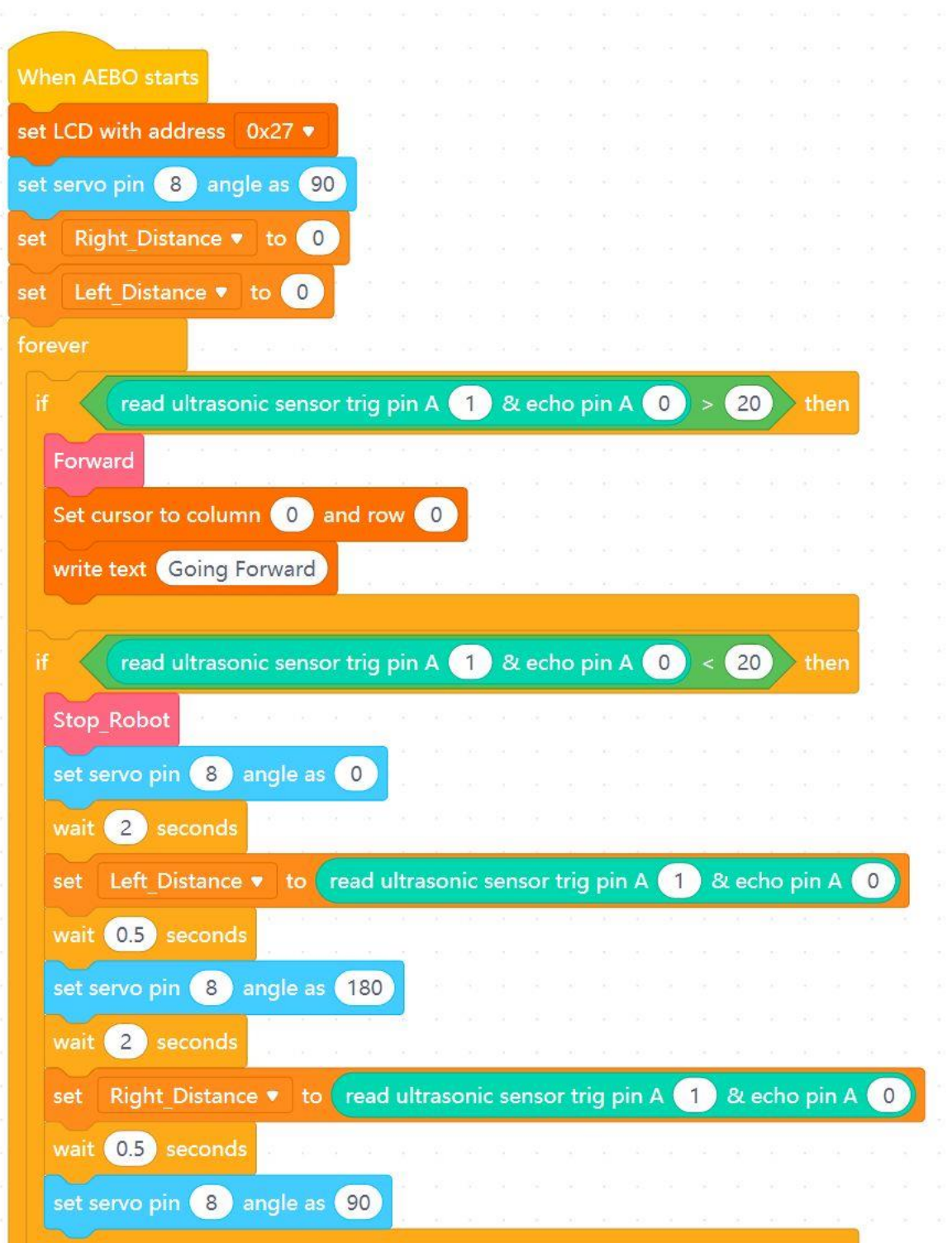


Figure 1.31

Continuation image of path finding robot figure 1.31

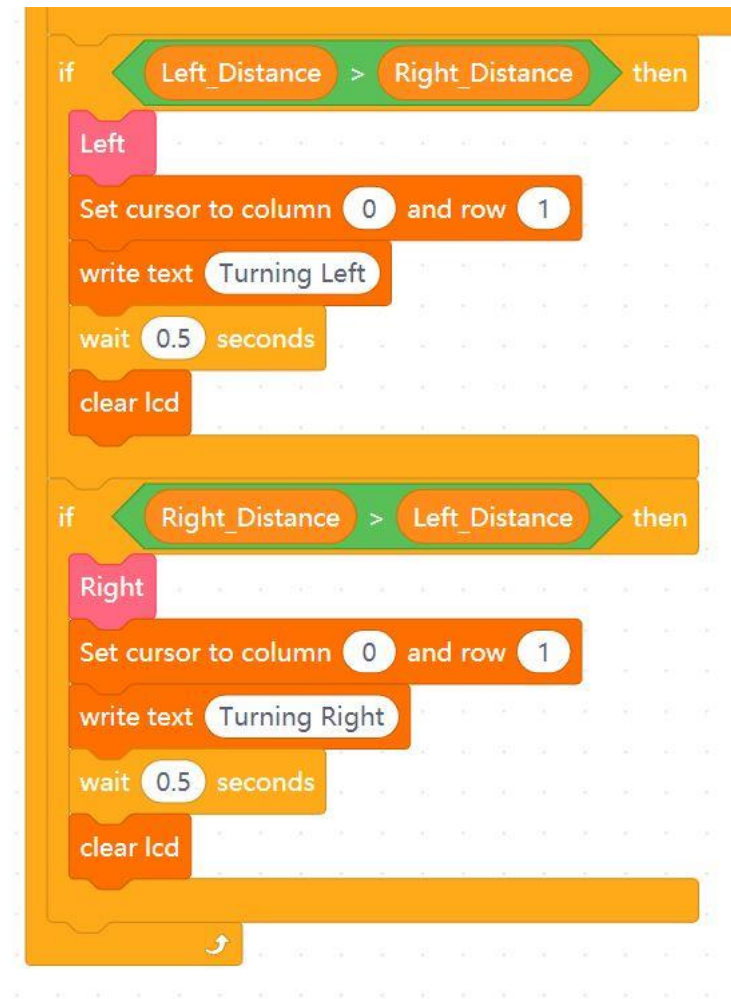


Figure 1.32